BANKING

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BANK TERM DEPOSIT SUBSCRIPTION PREDICTION USING MACHINE LEARNING

Problem statement:

Banks use **telephonic marketing campaigns** to promote term deposit subscriptions, but contacting every customer is time-consuming and expensive. The objective of this project is to analyze customer and campaign data to identify patterns influencing customer decisions and build a machine learning model that **predicts whether a client is likely to subscribe to a term deposit.** This helps banks improve marketing efficiency, reduce operational costs, and target potential customers more effectively.

PROJECT GOALS:

- Analyze customer banking data
- Perform EDA
- Identify important customer patterns
- Predict whether customer subscribes to term deposit
- Compare ML models

DATASET OVERVIEW:

The dataset used in this project is related to direct marketing campaigns conducted by a **Portuguese banking institution**. The campaigns were primarily carried out through phone calls to promote bank term deposits.

- Total Records: 45,211 customer entries
- Total Features: 18 attributes
- Dataset Type: Structured CSV dataset
- Target Variable: y
  "yes" → Customer subscribed to term deposit
  "no" → Customer did not subscribe

Customer Information: age, job, marital status, education
Financial Details: balance, housing loan, personal loan
Campaign Information: contact type, duration, campaign contacts
Previous Campaign Data: previous contacts, previous outcomes

WORKFLOW:

Dataset

↓

Data Cleaning

↓

Exploratory Data Analysis

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Feature Encoding

↓

Model Training

↓

Evaluation

↓

Prediction

LIBRARIES USED :

Python

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

DATA CLEANING:

NULL VALUES/MISSING DATA:

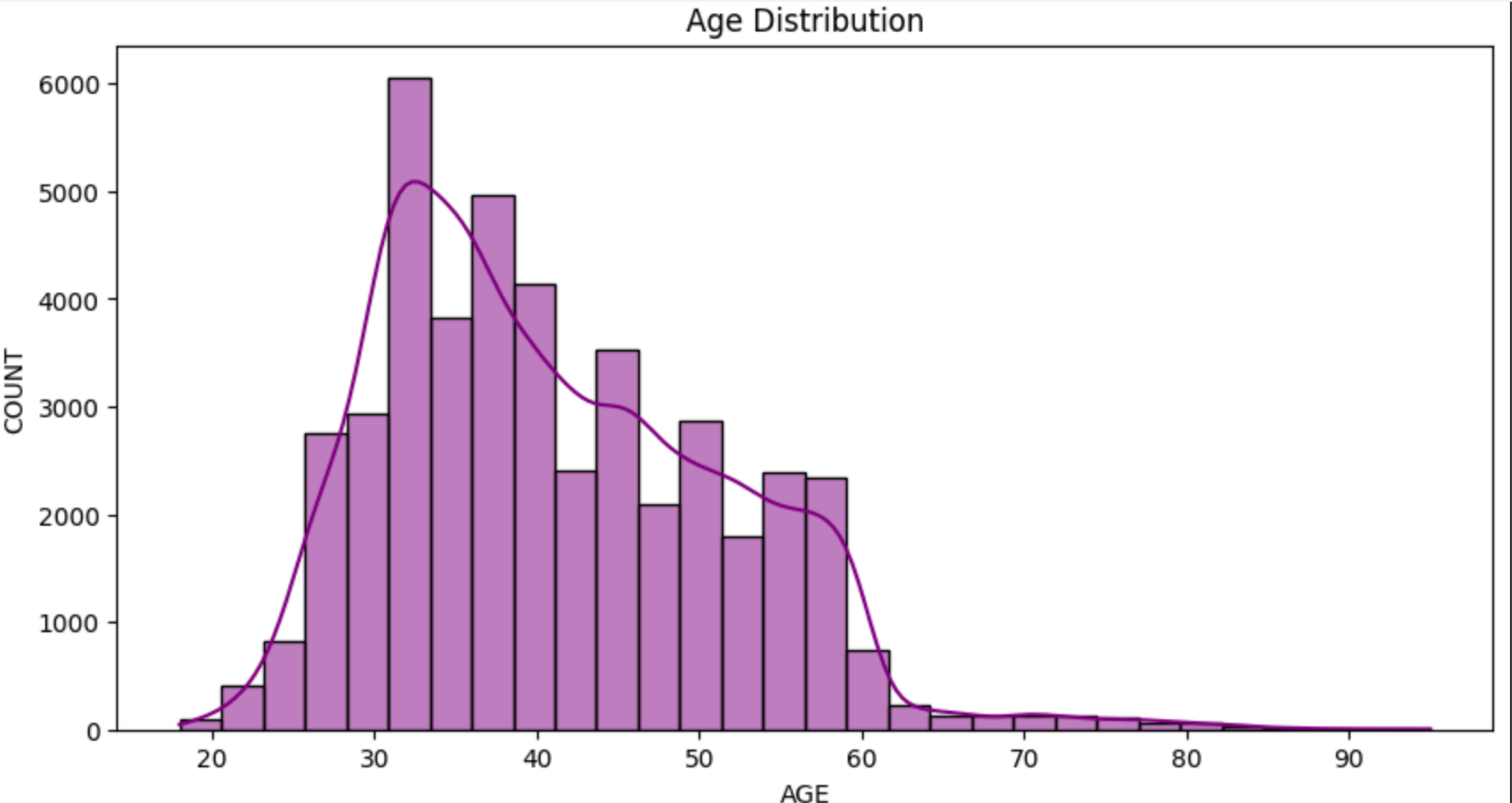
- 3 null values in Marital status and Education
- data.isnull().sum()

DUPLICATE VALUES:

- 5 duplicate values
- data.duplicated().sum()
  np.int64(5)
- data = data.drop_duplicates() - removes the duplicate data

EXPLORATORY DATA ANALYSIS (EDA)

AGE DISTRIBUTION

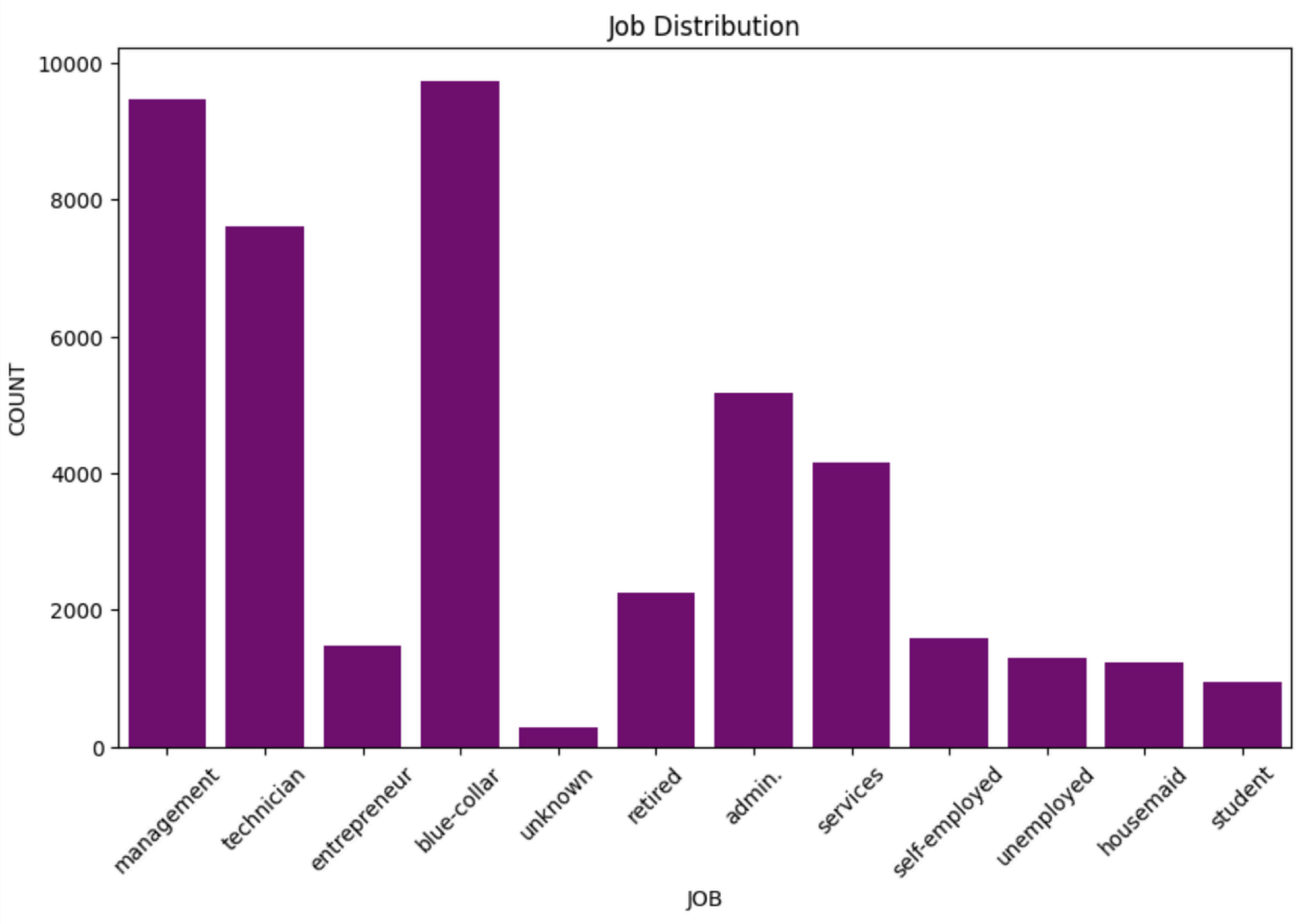


- The age distribution analysis shows that the **average age** of the clients is approximately **41 years**, indicating that most customers belong to the middle-aged working population. The **standard deviation of 10.62** suggests that the ages of most clients vary within around 10 years from the mean age, showing a moderate spread in the dataset.

- The **youngest client in the dataset is 18 years old**, while the **oldest client is 95 years old**, indicating a wide age range among customers. Additionally, **25% of the clients are below 33 years of age, 50% are below 39 years (median age), and 75% are below 48 years.** This indicates that a large proportion of the bank's customers fall within the **economically active age group of 30–50 years**.

- Overall, the distribution suggests that the **bank primarily targets middle-aged** individuals who are more likely to have stable income sources and investment capabilities, making them potential subscribers for term deposit schemes.

JOB DISTRIBUTION

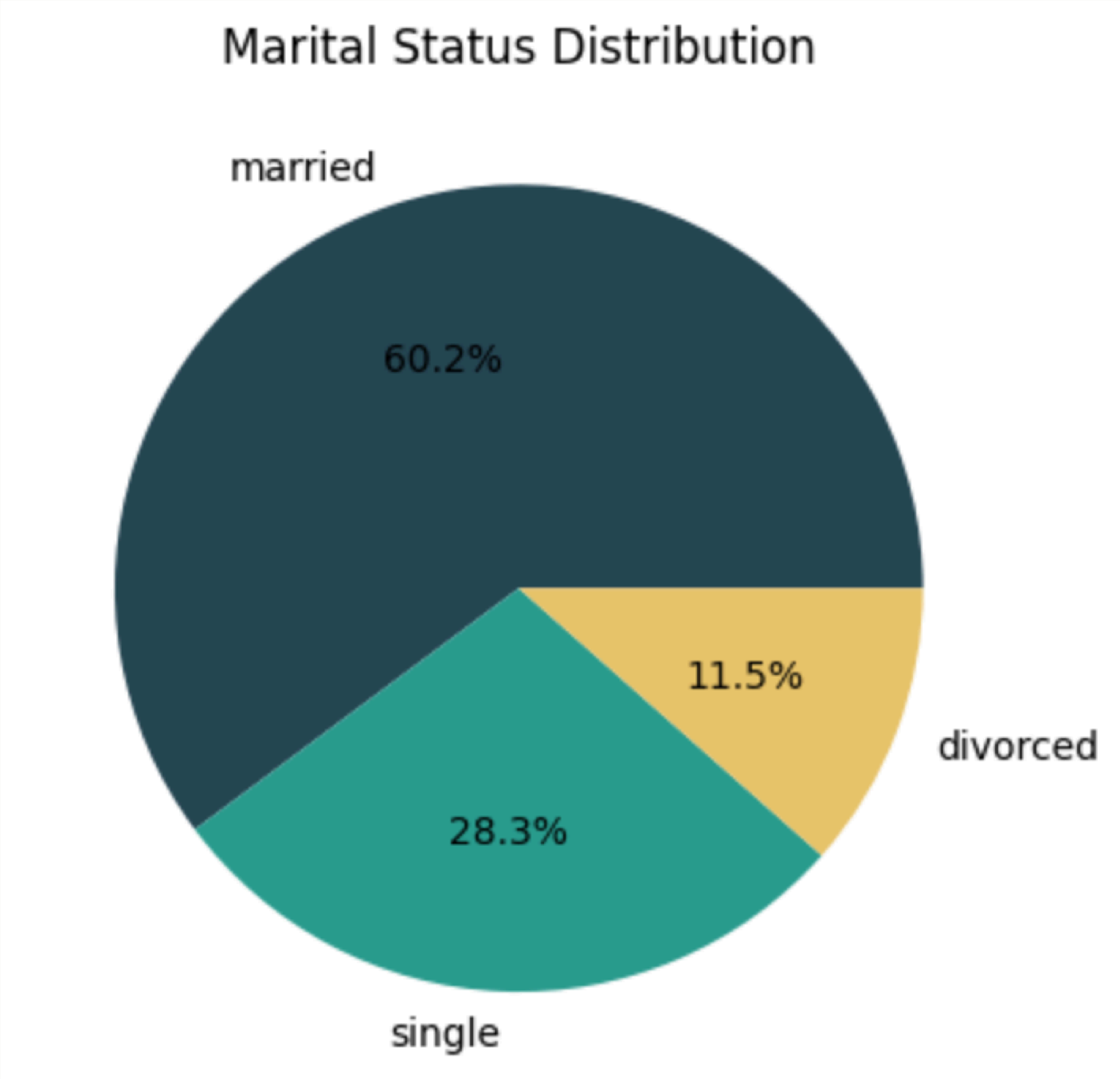


- The job distribution analysis shows that the **majority of clients belong to the blue-collar category with 9,732 customers**, accounting for approximately **21.5%** of the dataset. This is followed by **management professionals with 9,458 customers (20.9%) and technicians with 7,597 customers (16.8%)**. **Administrative staff** also represent a significant portion with **5,171 clients (11.4%)**.

- **Service workers account for around 9.2%** of the customers, while **retired individuals represent 5.0%** of the dataset. **Self-employed clients (3.5%), entrepreneurs (3.3%), unemployed individuals (2.9%), and housemaids (2.7%)** make up smaller portions of the customer base. **Students** account for approximately **2.1%** of the dataset, and only **0.6%** of the records belong to customers with **unknown job categories**.

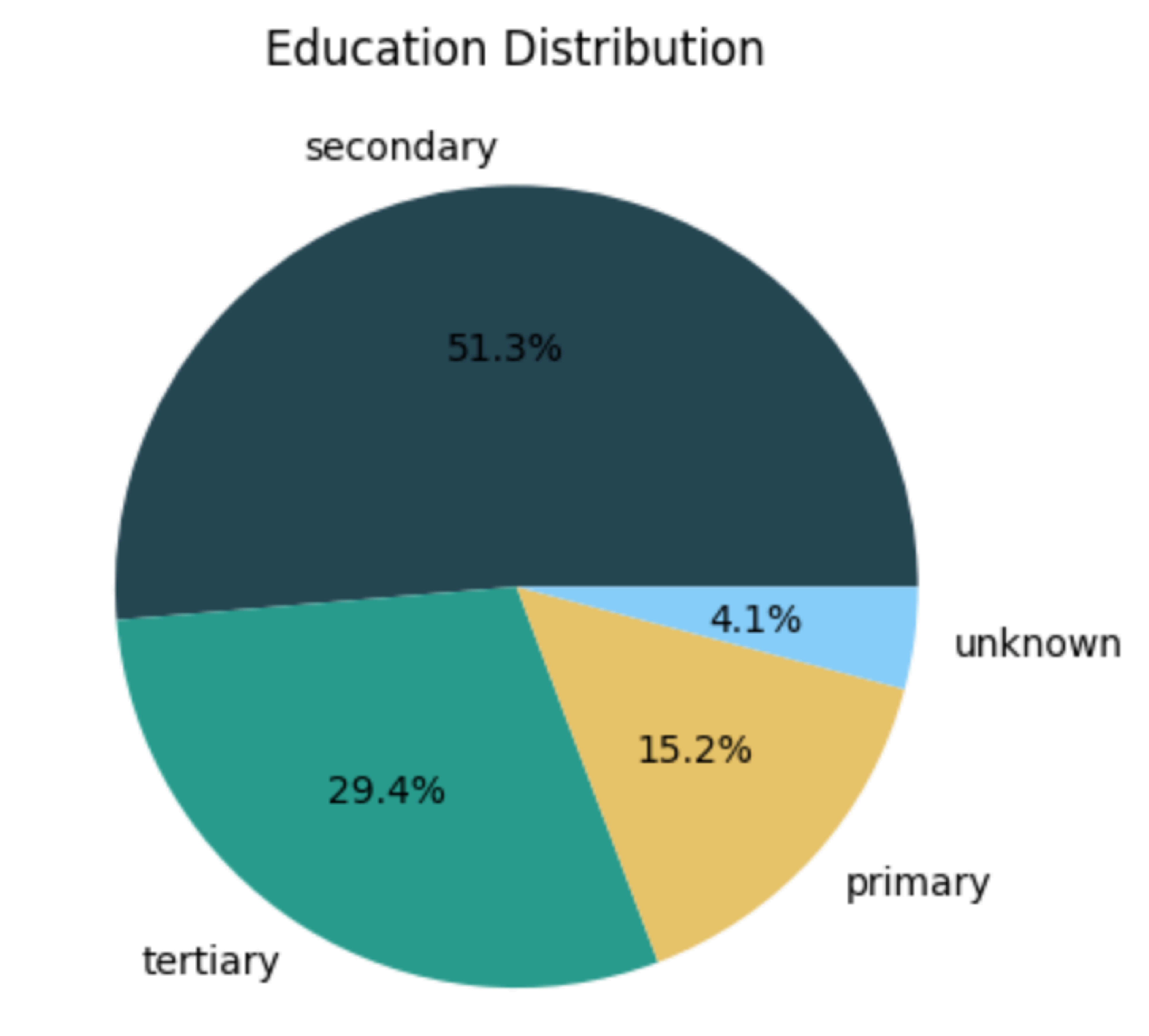
- Overall, the analysis indicates that the bank's marketing campaigns **primarily target working professionals, skilled workers, and management employees, who are more likely to possess stable income and investment potential for term deposit subscriptions**.

MARITAL STATUS



- The marital status distribution analysis shows that the majority of clients are **married, accounting for approximately 60.2%** of the dataset. **Divorced clients represent around 28.3%**, while **single clients make up the remaining 11.5%** of the customer base.
- This distribution indicates that the bank's marketing campaigns **primarily target married individuals, who may have greater financial responsibilities, stable income sources, and long-term investment interests such as term deposits**. The comparatively lower proportion of single clients suggests that middle-aged and family-oriented customers form a significant segment of the bank's customer base.

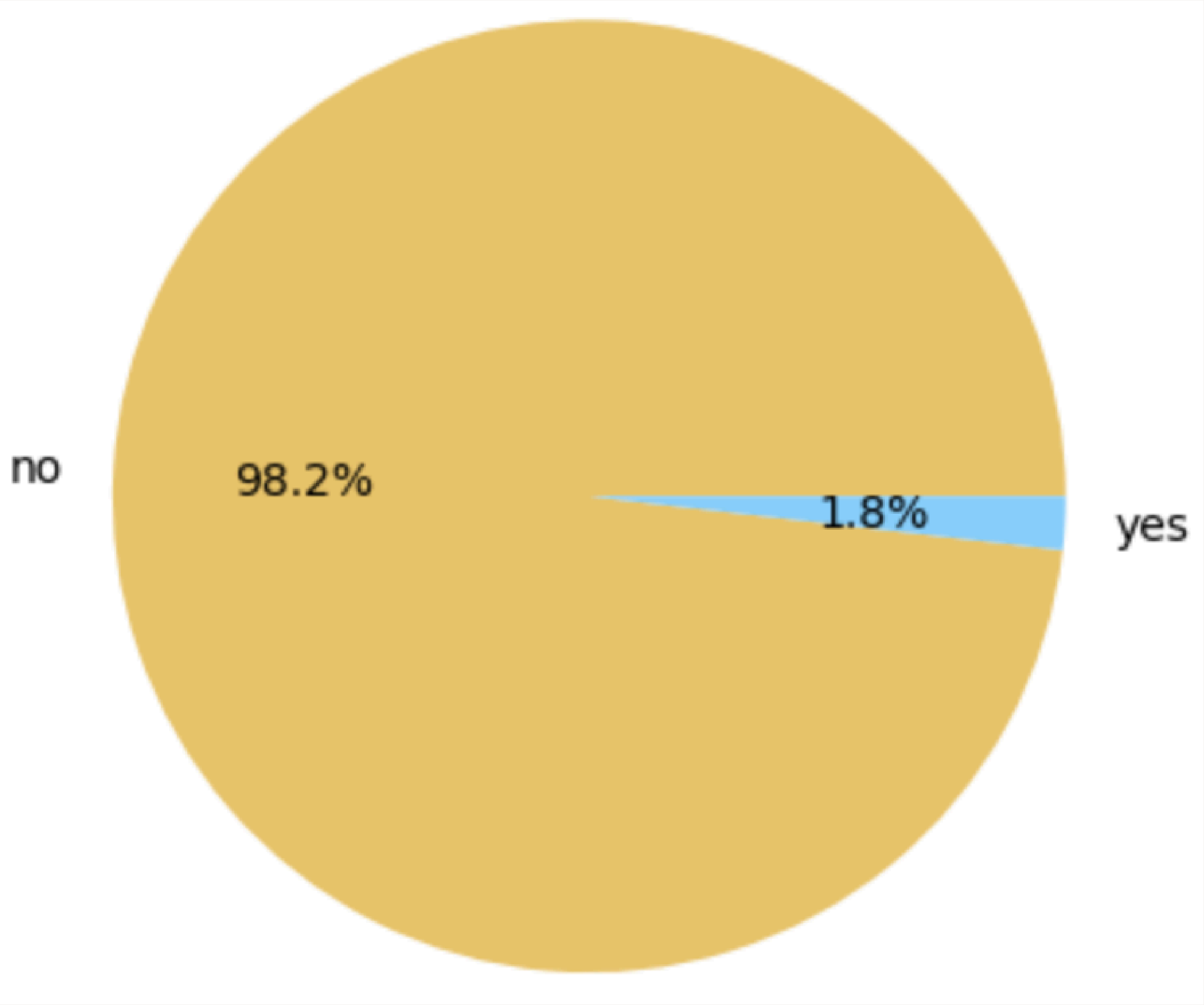
EDUCATION DISTRIBUTION



- The education distribution analysis shows that the **majority of clients have secondary education, accounting for approximately 51.3%** of the dataset. Customers with **tertiary education represent around 29.4%**, while clients with **primary education make up 15.2%** of the total customer base. Only **4.1% of the records belong to customers with unknown educational backgrounds**.

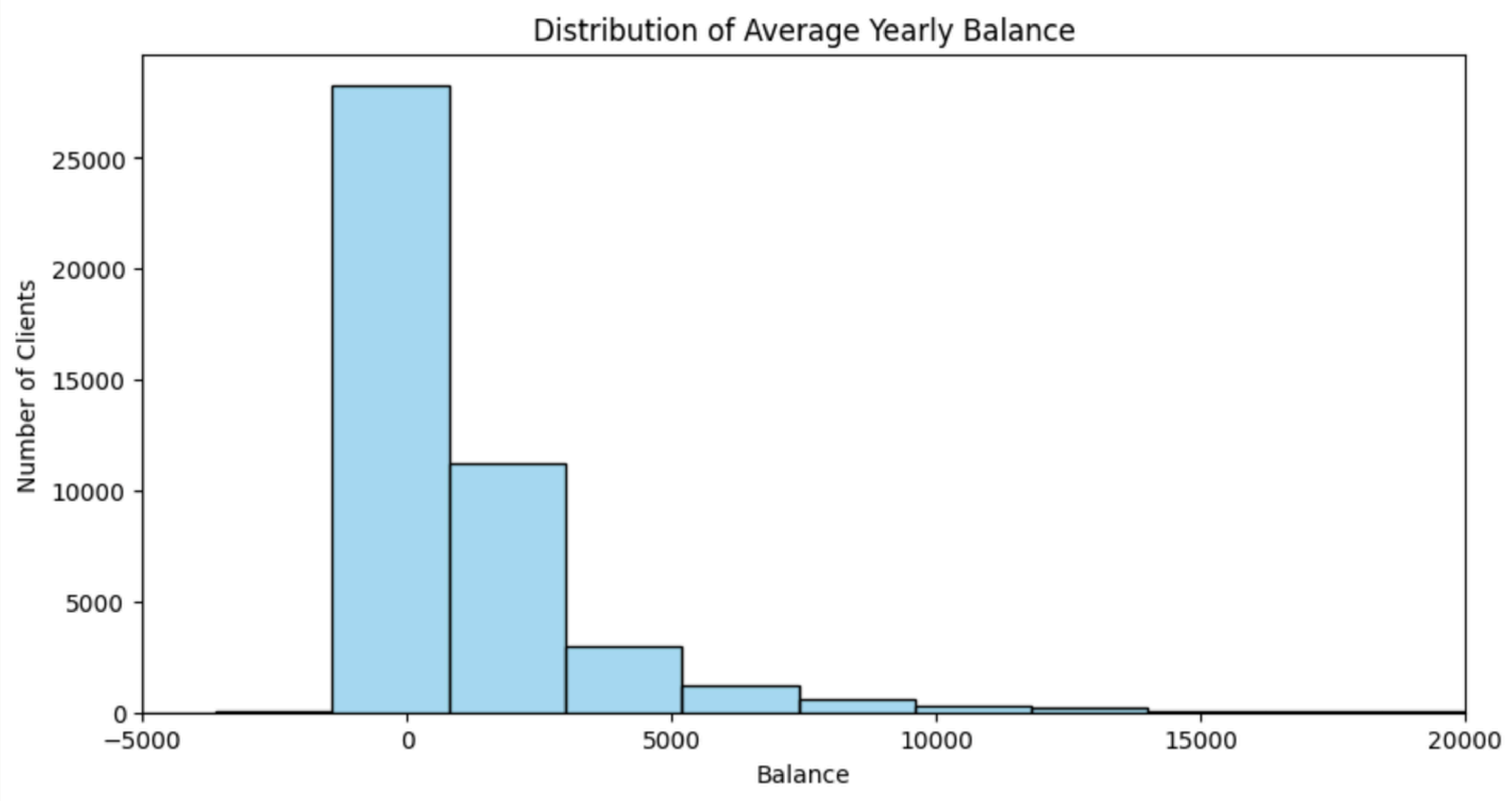
- This distribution indicates that **most customers possess at least a basic or intermediate level of education, with a significant proportion also having higher educational qualifications**. The large share of secondary and tertiary educated clients suggests that the bank's marketing campaigns primarily **target financially aware and economically active individuals** who may be more likely to understand and invest in banking products such as term deposits.

DEFAULT DISTRIBUTION



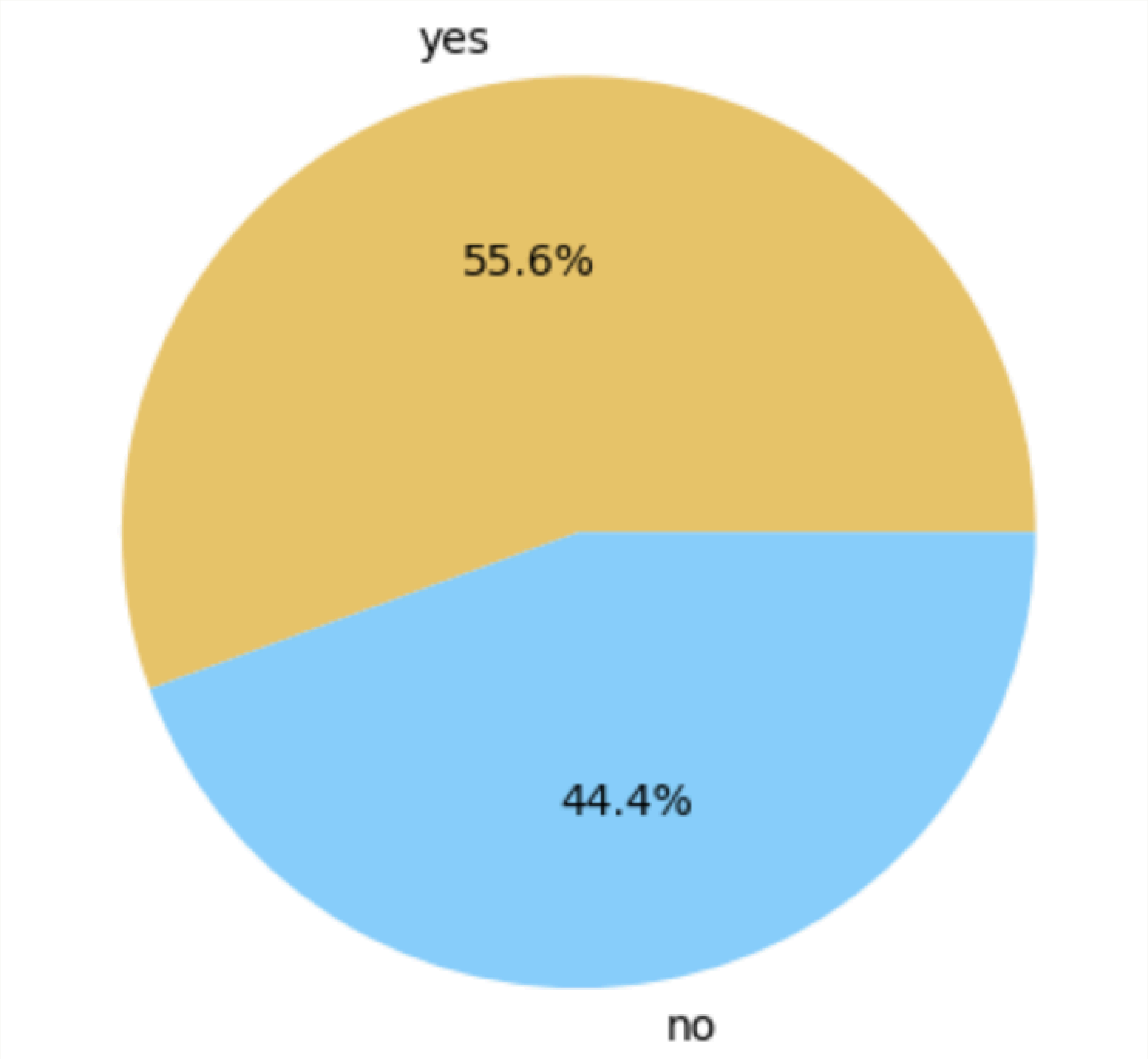
- In banking, “credit in default” means a customer has failed to repay a loan or credit obligation on time.
- The analysis of credit default status shows that the vast **majority of clients, approximately 98.2%, do not have credit in default, while only 1.8% of customers have defaulted credit records**.
- This indicates that **most customers in the dataset maintain stable financial behaviour and good credit standing**. The very small proportion of default cases suggests that the bank primarily deals with financially reliable customers, which may positively influence the likelihood of term deposit subscriptions and reduce financial risk during marketing campaigns.

DISTRIBUTION OF AVERAGE YEARLY BALANCE



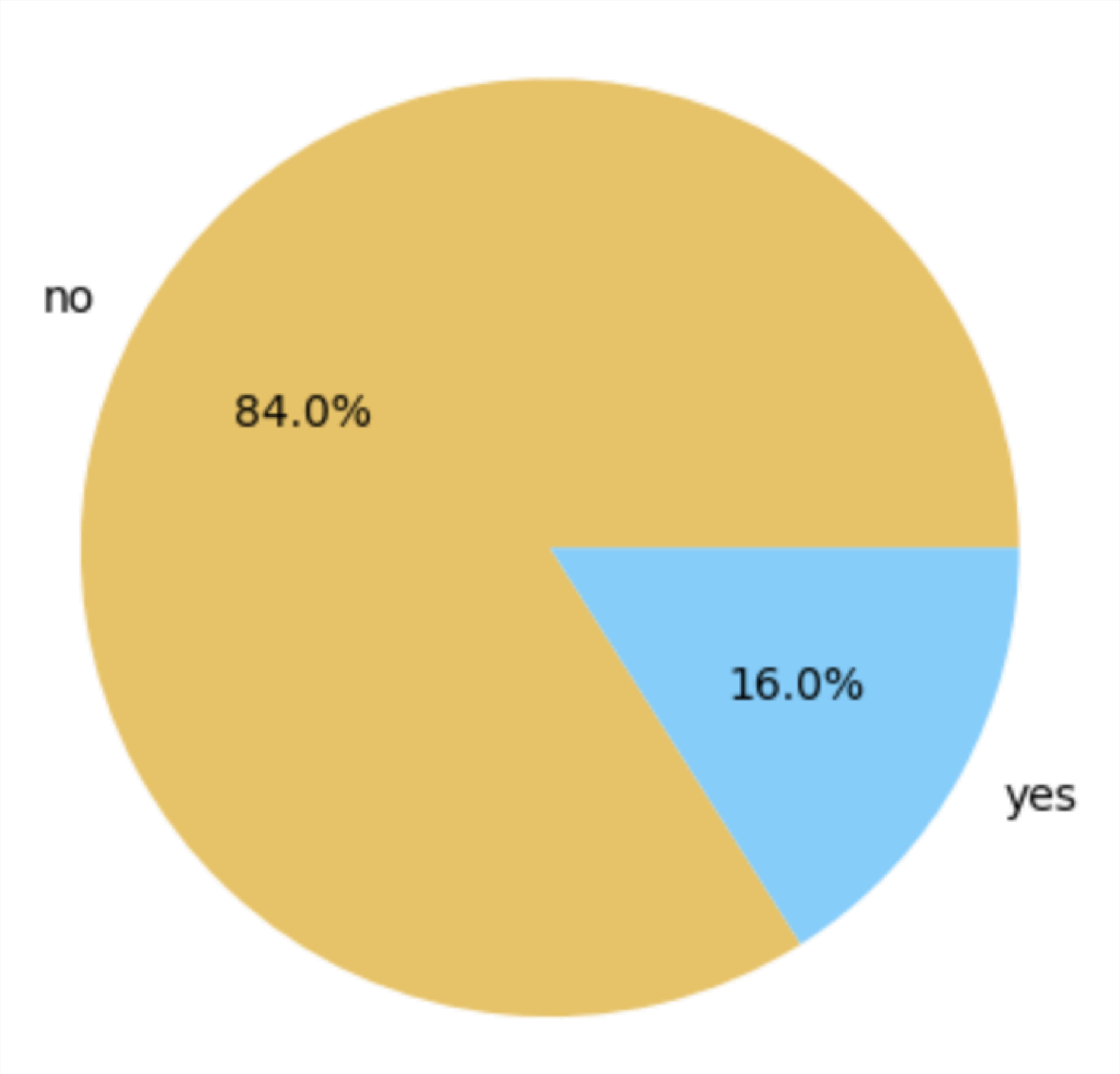
The balance distribution is **positively skewed**, indicating that **most customers maintain low to moderate account balances**, while a small number of customers have very high balances. The presence of outliers shows significant variation in customer financial status, which may influence term deposit subscription behavior.

HOUSING LOAN



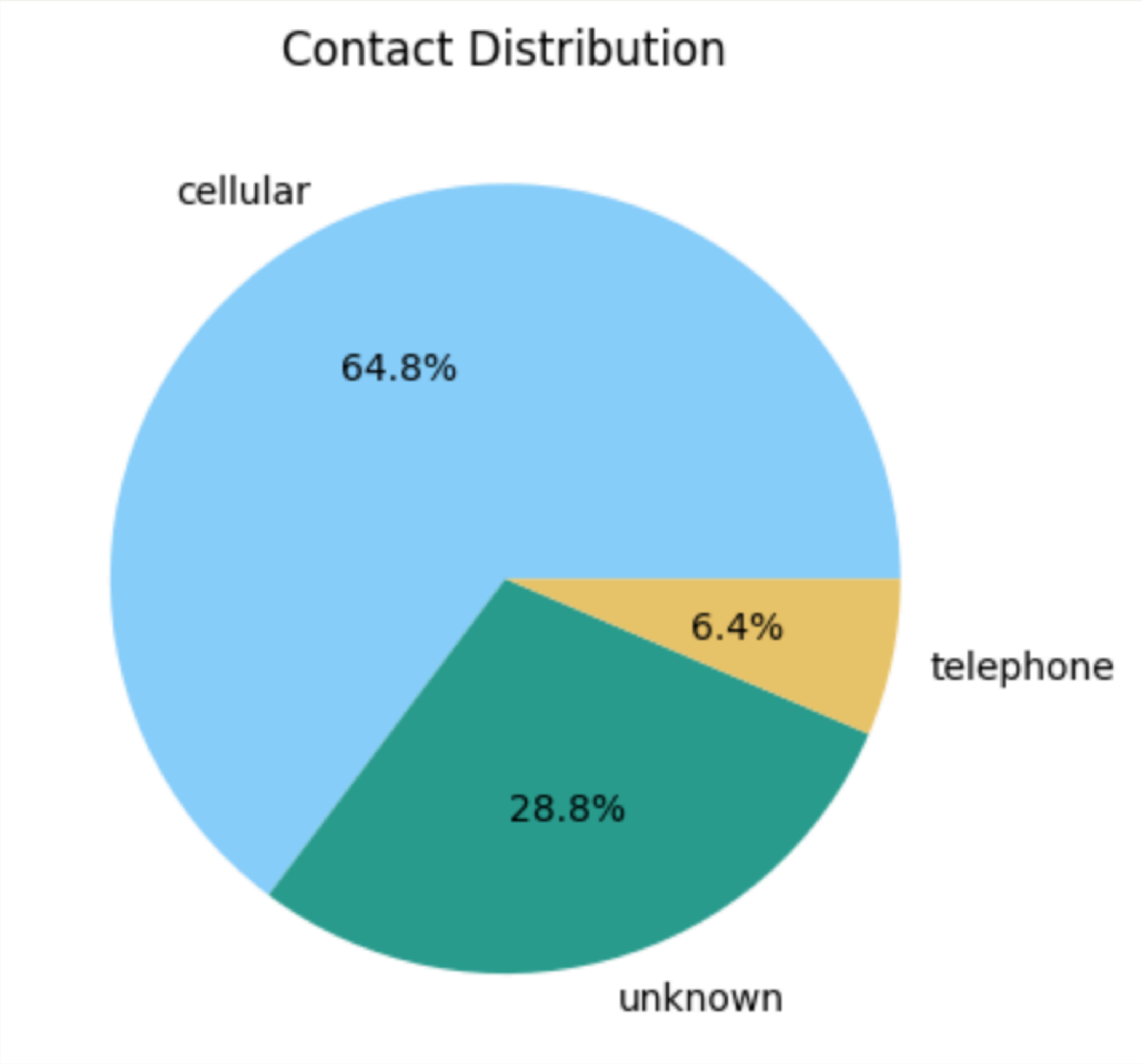
The housing loan analysis shows that **55.6% of customers have a housing loan, while 44.4% do not**. This indicates that more than half of the clients are managing long-term financial commitments, suggesting a customer base with stable income and active banking relationships.

PERSONAL LOAN



The personal loan analysis reveals that **84.0% of customers do not have personal loans, while only 16.0% possess personal loans**. This suggests that the **majority of clients have lower short-term debt obligations, which may positively influence their ability to invest in banking products such as term deposits.**

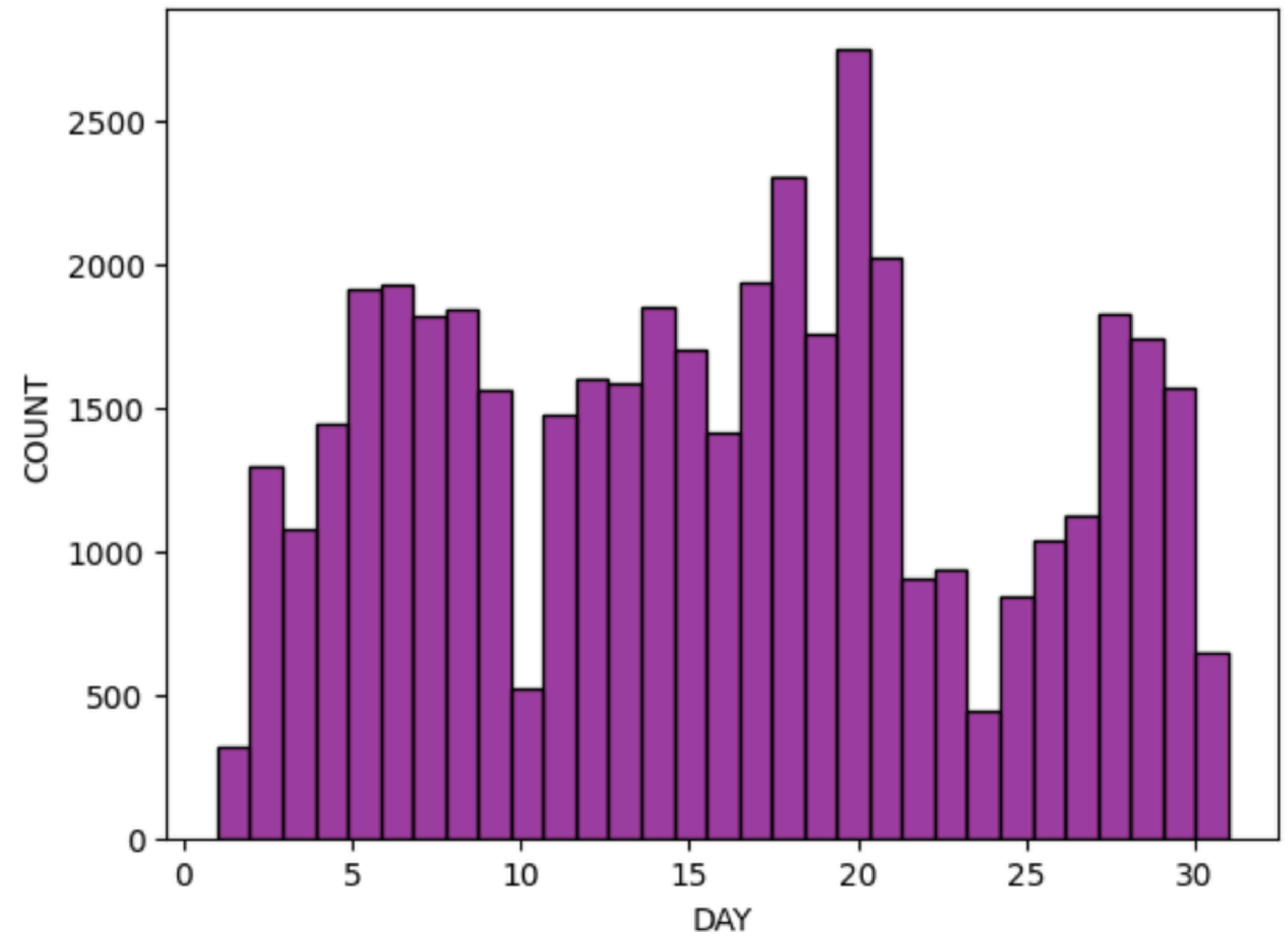
CONTACT DISTRIBUTIUON



- The contact distribution analysis shows that **cellular communication is the most commonly used contact method, accounting for 64.8%** of customer interactions. Contacts with unknown communication type represent 28.8% of the dataset, while **telephone communication accounts for only 6.4%**.

- This indicates that **mobile-based communication** plays a major role in the bank's marketing campaigns, likely due to its **higher accessibility and effectiveness in reaching customers**. The relatively low use of telephone contact suggests a shift toward more convenient and direct communication channels such as cellular networks.

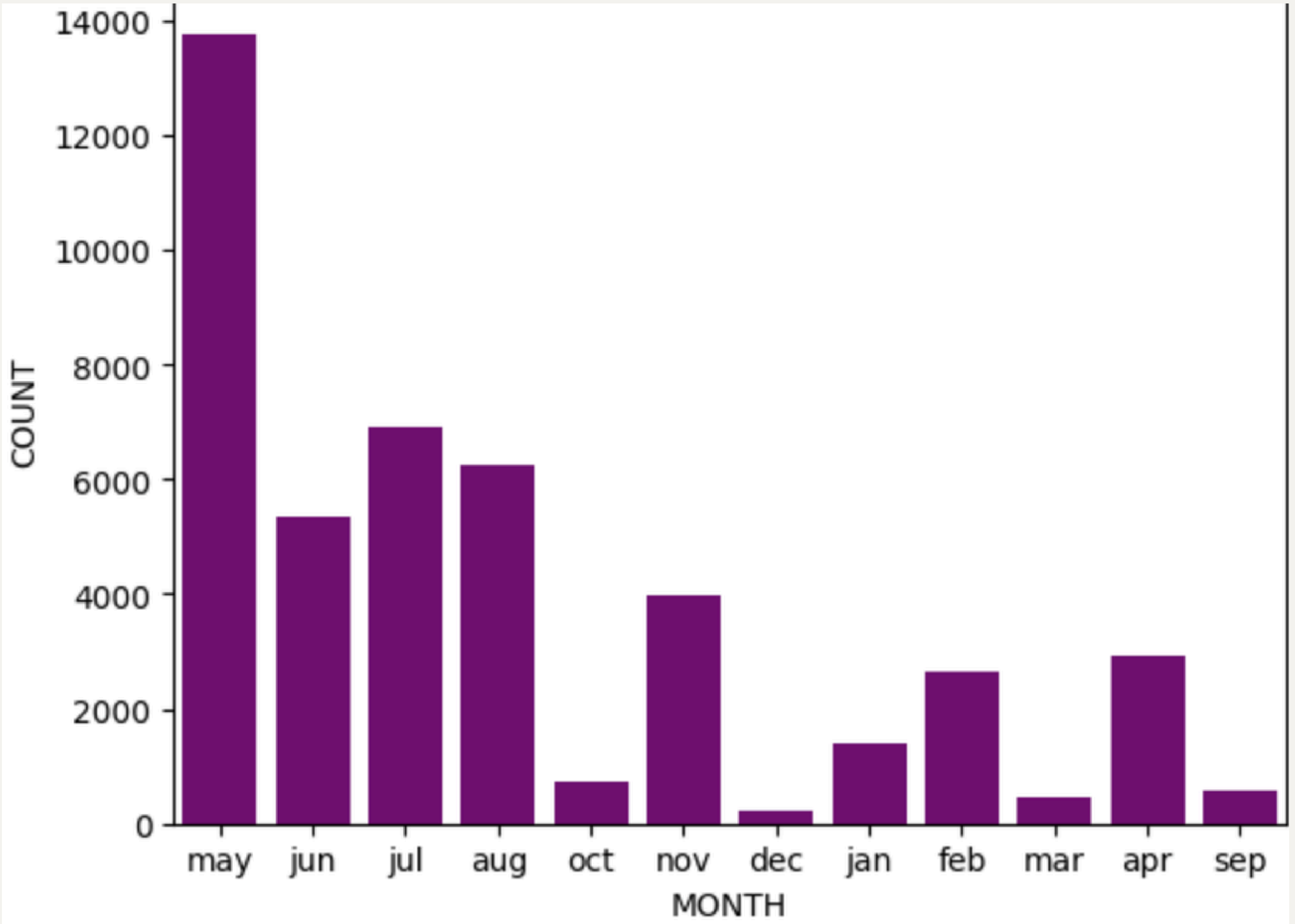
LAST CONTACT DAY OF THE MONTH



- The distribution of the last contact day of the month shows that **customer contacts were spread throughout the month**, with **higher contact frequencies observed during the middle and later days of the month**. The highest number of contacts occurred around the 20th day, while comparatively fewer contacts were made during the beginning of the month.

- This indicates that the **bank conducted marketing campaigns consistently across different days**, with **certain periods showing increased customer engagement activity**.

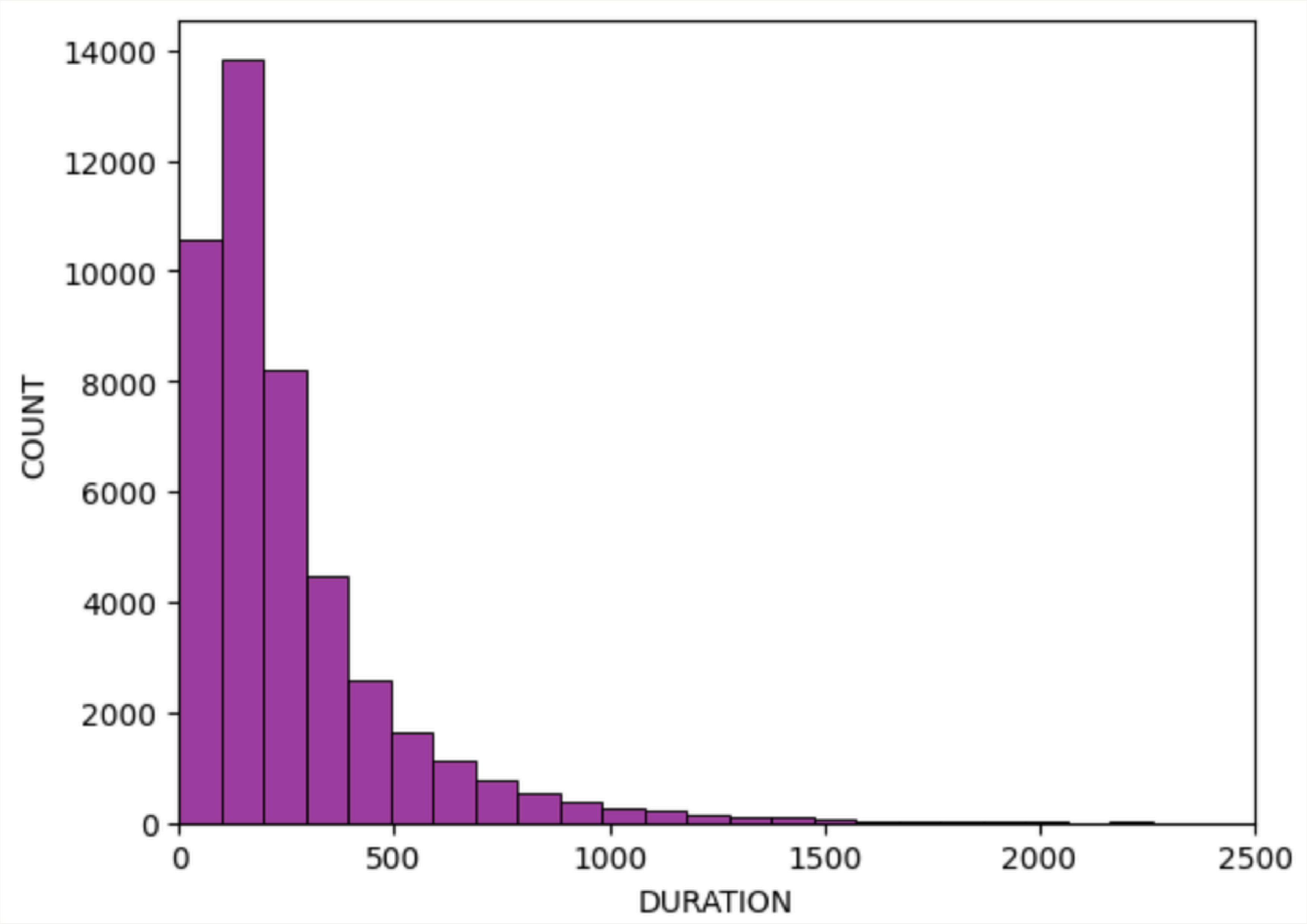
LAST CONTACT MONTH



- The distribution of the last contact month shows that the **majority of customer interactions took place in May**, making it the most active month for marketing campaigns. July and August also recorded a high number of contacts, followed by June and November.

- On the other hand, months such as December, March, and September had significantly fewer customer interactions. This uneven distribution indicates that the **bank focused its telemarketing campaigns more heavily during certain months of the year, possibly to improve customer engagement and increase the chances of term deposit subscriptions.**

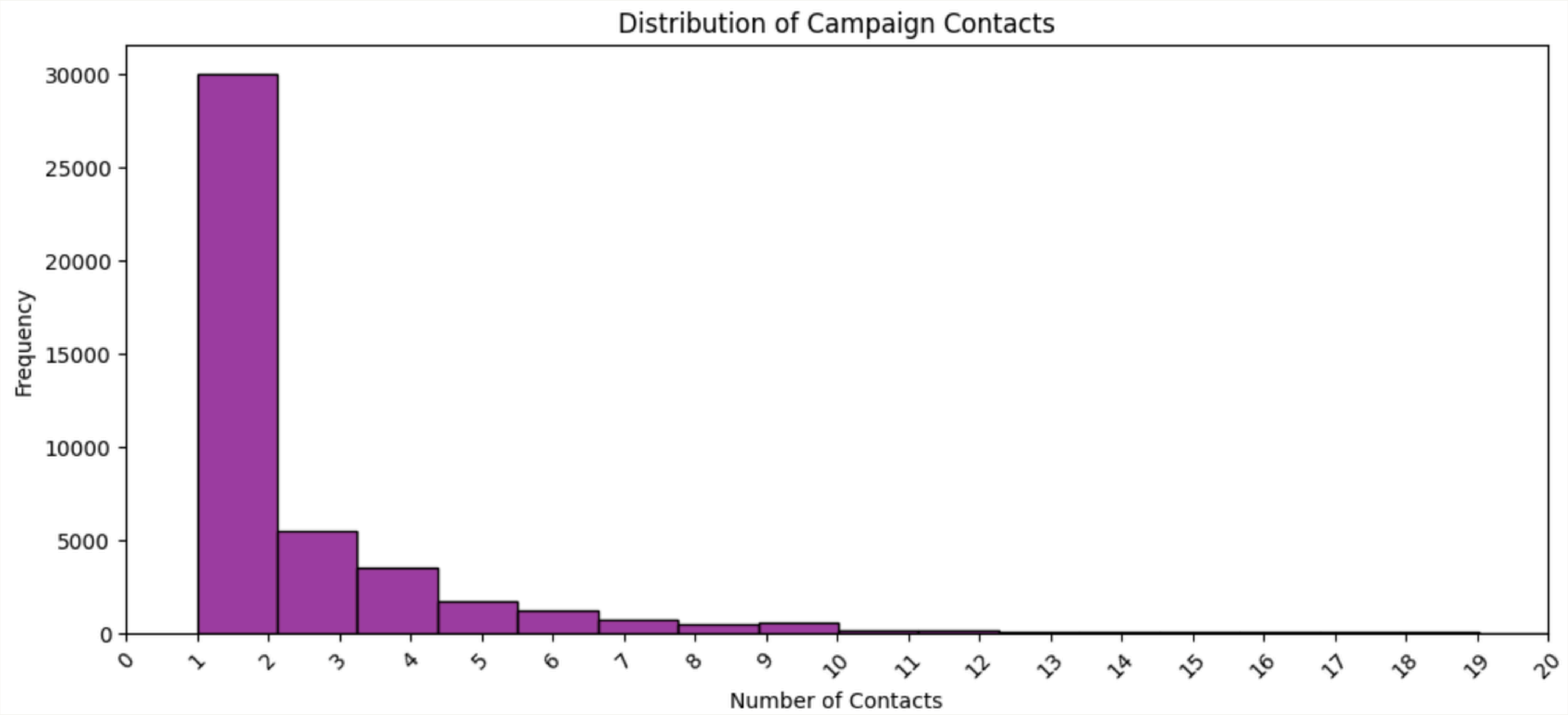
DISTRIBUTION OF DURATION OF CALL



- The call duration distribution is **positively skewed**, indicating that **most calls had shorter durations**, while only **a few calls lasted for a very long time**. The **most frequent call durations were around 90–125 seconds**, whereas extreme durations above 1000 seconds were rare.

- This variation suggests that customer engagement levels differed across interactions, with longer calls potentially reflecting greater interest in term deposit subscriptions.

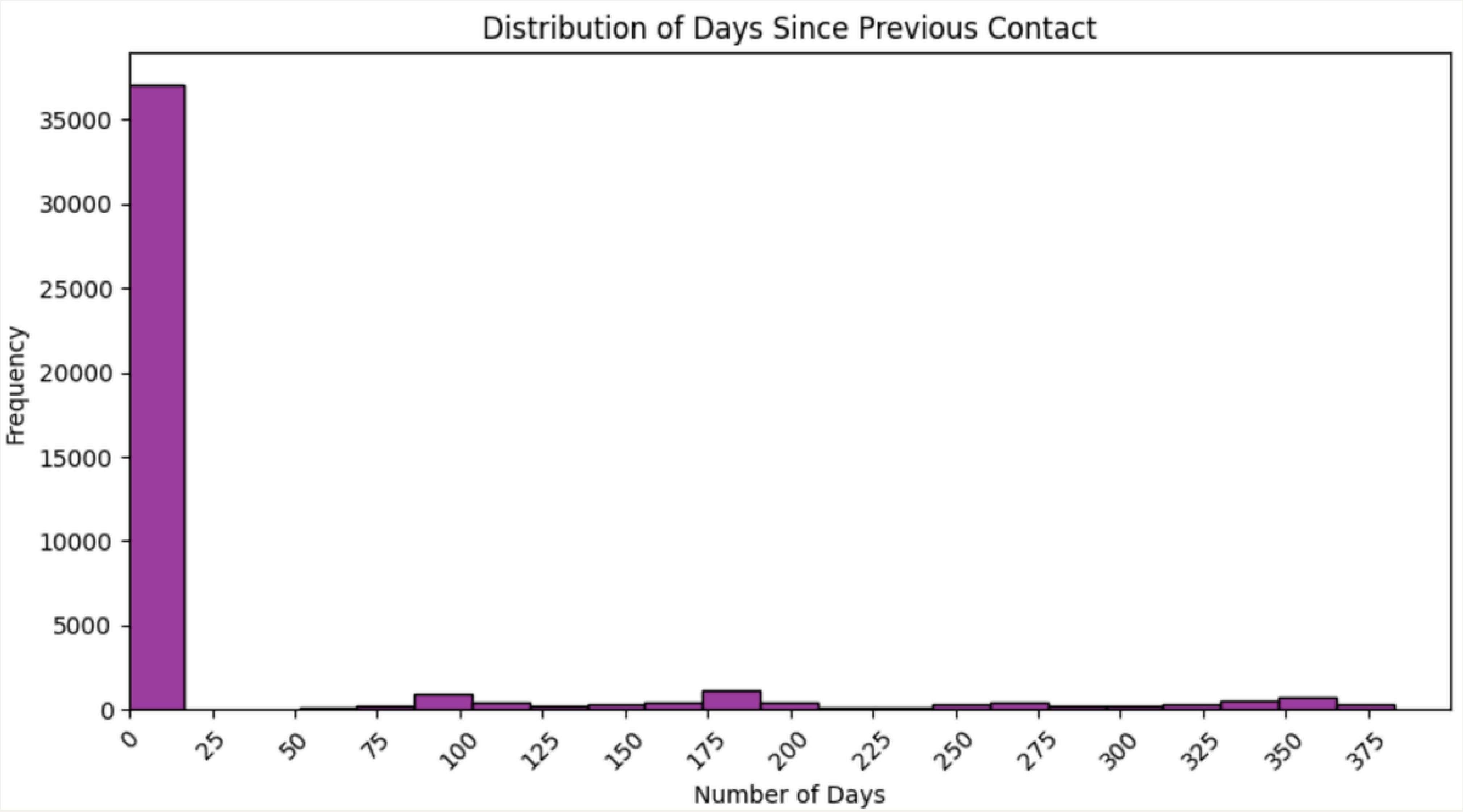
NUMBER OF CONTACTS



- The analysis shows that the majority of clients were contacted only a few times during the marketing campaign. Around **17,544 customers were contacted once, while 12,505 customers were contacted twice and 5,521 customers were contacted three times**. The number of clients decreases significantly as the contact frequency increases, with only a very small number of customers being contacted more than 20 times.

- This indicates that the **bank generally avoided excessive repeated calls in order to prevent spam-like interactions and customer inconvenience.** Additional follow-up contacts were likely used selectively for customers who showed higher potential interest in term deposit subscriptions.

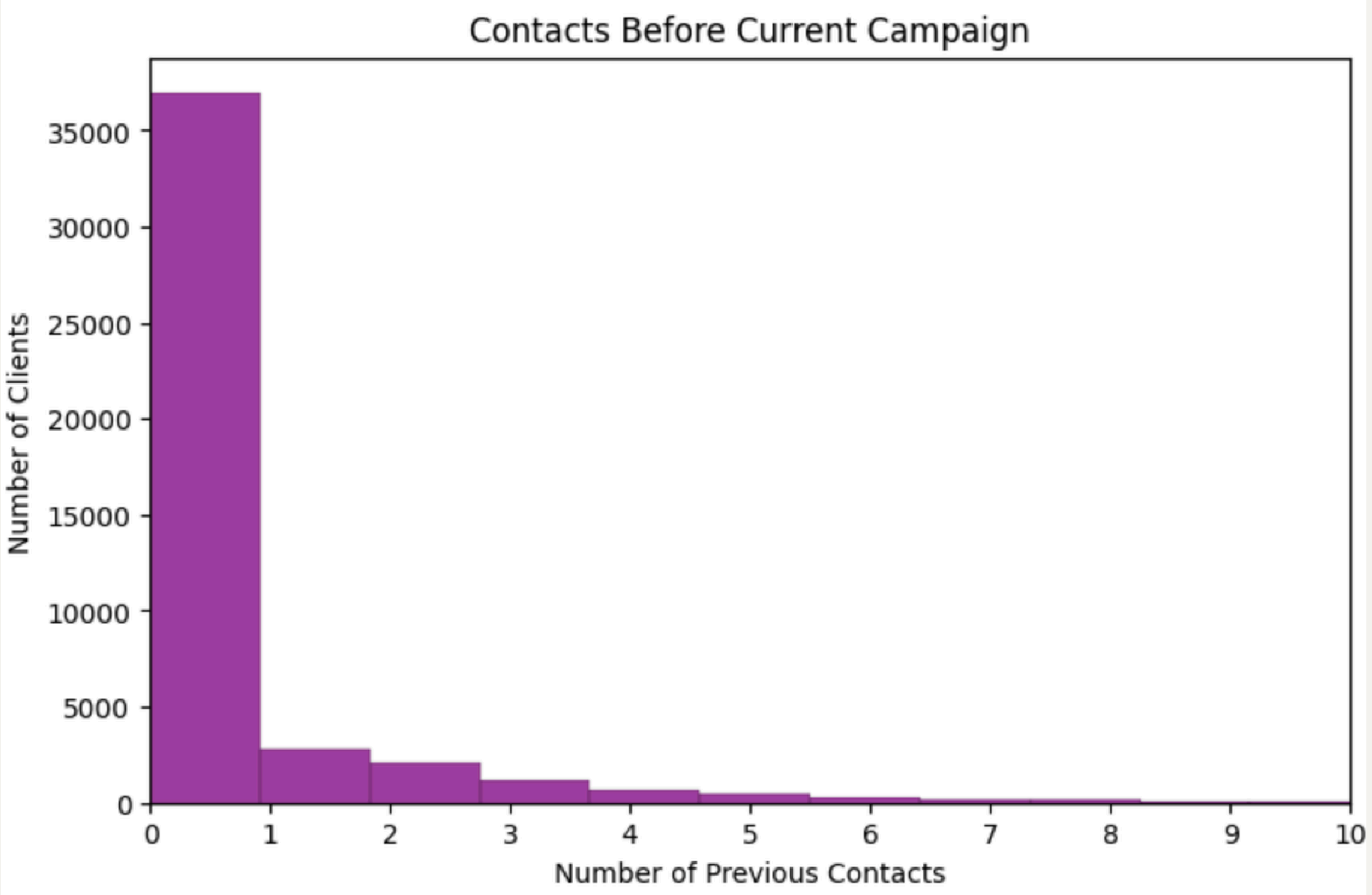
DAYS SINCE PREVIOUS CONTACT



- The pdays column shows the number of days passed since the client was last contacted in a previous campaign. From the graph, **most clients are concentrated near 0 / -1, which means the majority of clients were not contacted previously.** Only a small number of clients had previous contact records, with values spread between around 50 to 375 days.

- This shows that the dataset is **highly skewed because most clients had no previous campaign contact**, while only a few clients were contacted again after many days.

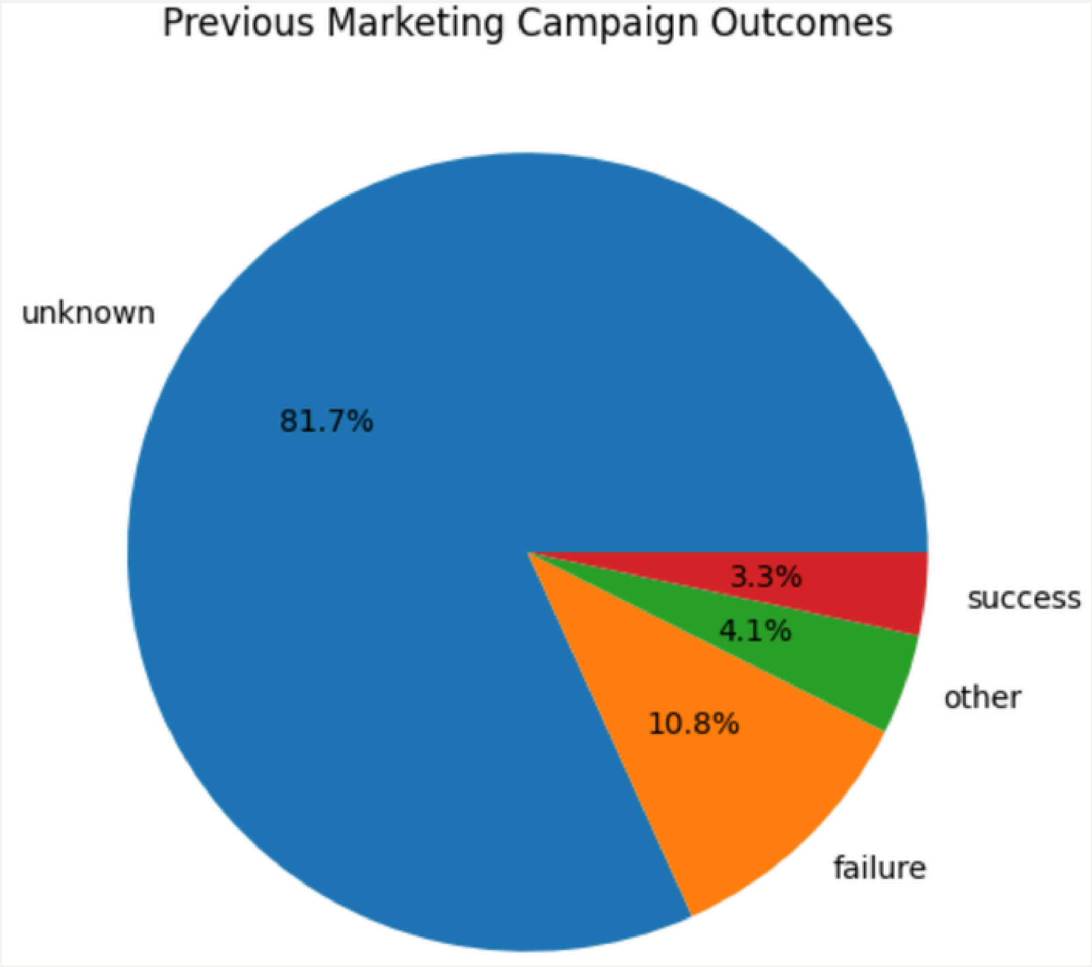
NUMBER OF PREVIOUS CONTACTS



- The previous column shows how many times each client was contacted before the current campaign. **Most clients had 0 previous contacts, meaning they were contacted for the first time** in this campaign. Only a small number of clients had been contacted one or more times before.

- This means the distribution is **highly skewed toward 0**. Previous campaign contact history is available for only a limited number of clients. Therefore, most clients in the dataset were new contacts for the current marketing campaign.

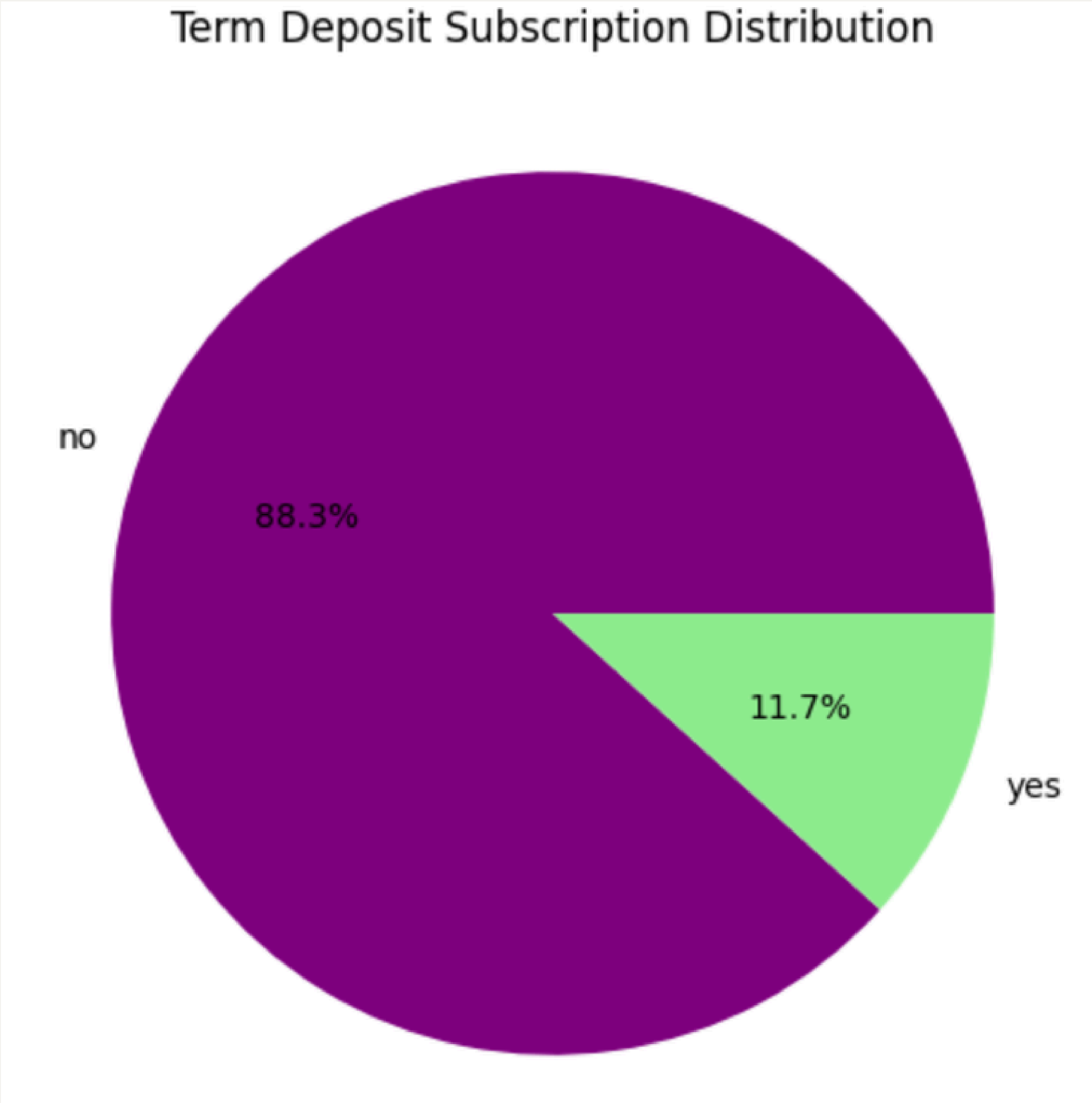
PREVIOUS CAMPAIGN OUTCOMES



- The poutcome column shows the result of the previous marketing campaign. **Most clients have the value unknown, which means there was no clear previous campaign result or the client was not previously contacted.** Among the known outcomes, **failure is more common than success**, while other appears for a smaller group of clients.

- This shows that previous campaign success was limited, and for most clients, previous campaign information is not available. The success category can still be important because clients who responded positively in a previous campaign may have a higher chance of subscribing again.

TERM DEPOSIT SUBSCRIPTION



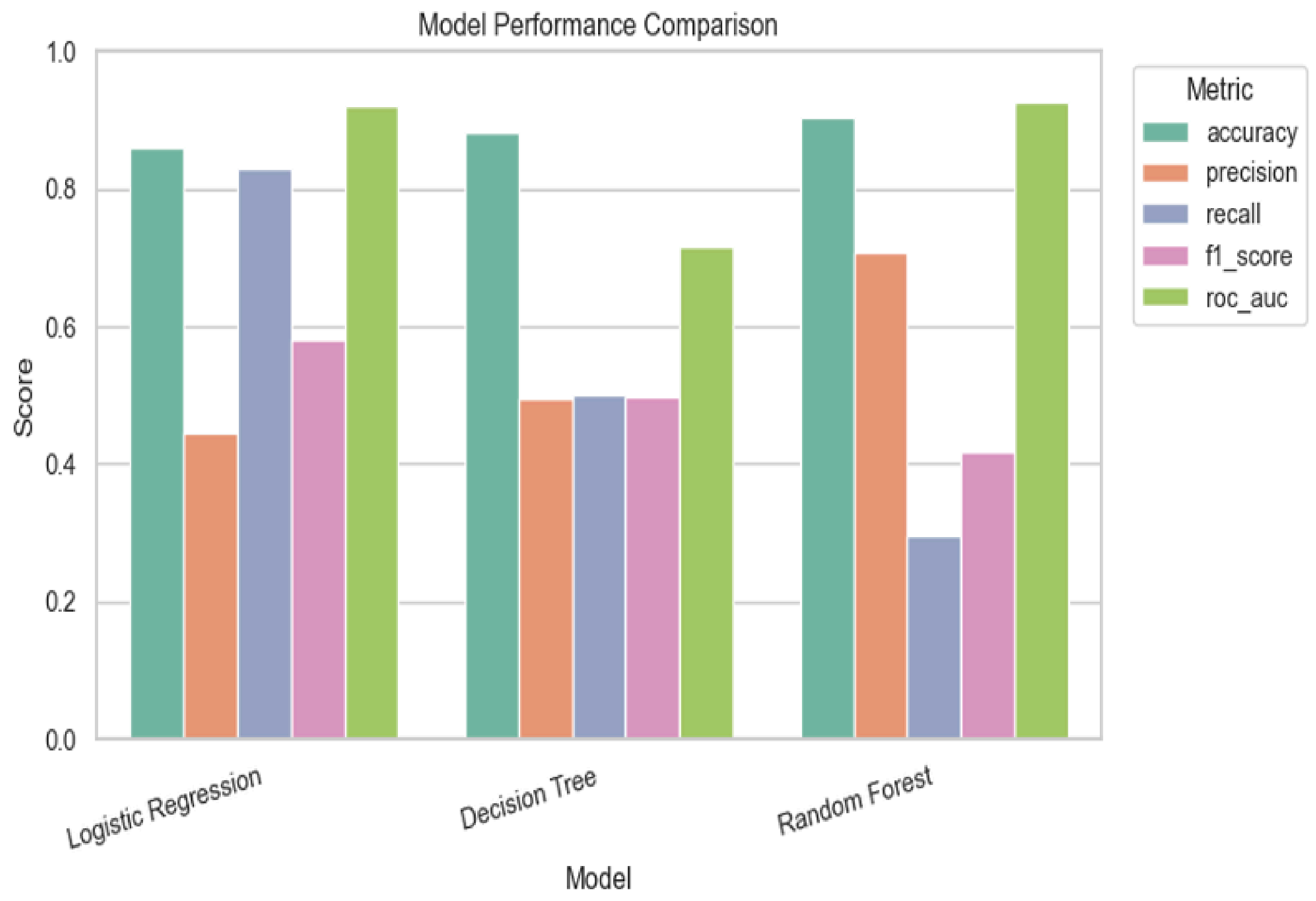
- This means only about **1 out of every 9 clients subscribed to the term deposit**.

- This indicates that the marketing campaign had a **low conversion rate**. The dataset is also **highly imbalanced** because the no class is much larger than the yes class. Because of this, **accuracy alone is not enough to judge the machine learning model**. For example, a model could predict no for most clients and still get high accuracy. So, metrics like precision, recall, F1 score, and ROC-AUC are more useful for evaluating model performance.

Model Comparison

Logistic Regression delivered the best F1 score, while Random Forest had the highest accuracy and ROC-AUC.

Performance comparison



Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.859	0.446	0.830	0.580
Decision Tree	0.882	0.494	0.500	0.497
Random Forest	0.903	0.708	0.296	0.417

Logistic Regression

Best model by F1

0.580

Best F1 score